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CMSC 204 Morse Code Project 5 Writeup

1. What did you learn?

I learned how to utilize a mix of 203 and 204 concepts (StringBuilder, string formatting, constructors, exceptions, HashMap, trees, nodes, etc.) to create a program that converts Morse code into English.

2. What issues did you encounter, if any?

I did not run into any major issues but I did notice some of the variable fields outlined in the Javadoc were not necessary for implementation and actually caused my implementation to run incorrectly.

3. What test cases did you run?

I thoroughly tested all function methods (outlined in the provided Javadoc + my additional helper private methods), every constructor, and a wide variety of edge/special cases I came up with. I also made sure to avoid retesting methods that were already covered by the public tests to avoid

4. What were the expected and actual outputs (in the JavaFX GUI) when you ran it?

I expect that the morse code will be converted correctly into English text, but I don't know what the "definite" expected output is because this was not given to me. I will provide the screenshots down below.

Daisy.txt

-- . . .- . / -- . / -.- -.- .. -. / .-- . .- / .. -.-

DaisyDaisy.txt

.. -- /- .-. ... / -. .-. .- -- .. -- / .- .-. .-. / ..- --- .- / - /
.-. --- / --- ..- / -. -- -- ..-

howDoILoveThee.txt

.... --- .- / -. --- / .. / .-. --- / - / .-. .- / -- . / -. --- ..-
- / - / .- .- -. -- ...

LoveLooksNot.txt

.-. --- / .-. --- --- -. ... / -. --- - / .- .. - / - / .- -- /
-- / .- .. - / - / -- .. -. --

Stringtest (NOT a .txt file, used to test conversion with a string instead of a file).

.-. .- -- - --- -. / .-. .-. --- -- .-. .- -- -- ..- .- --

**Expected output: "python programming"*

5. Will your project be able to pass a set of private test cases?

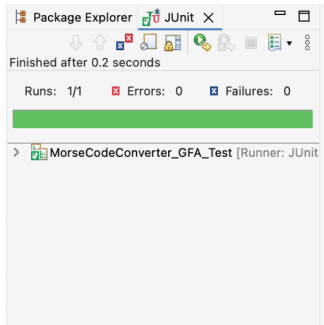
While I cannot say this for sure, I highly believe my project can pass a set of private test cases because I tested all my projects exactly the same in the way I mentioned for question #3 and so far all of my projects in CMSC 204 have passed all private test cases. I also did not hardcode any part of my project to produce the expected output artificially.

6. Assumptions

I assumed this project would be fairly short given the relatively small number of classes I was required to implement and having experience with a similar project in CMSC 203 that was based on encrypting/decrypting strings.

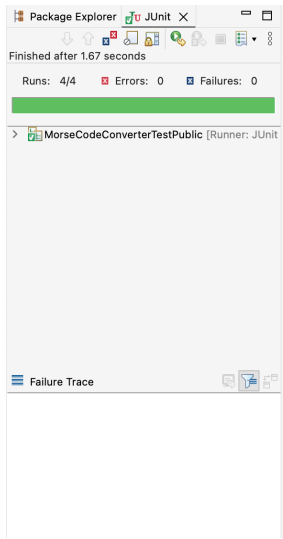
JUnit Test Screenshots:

MorseCodeConverterGFATest:



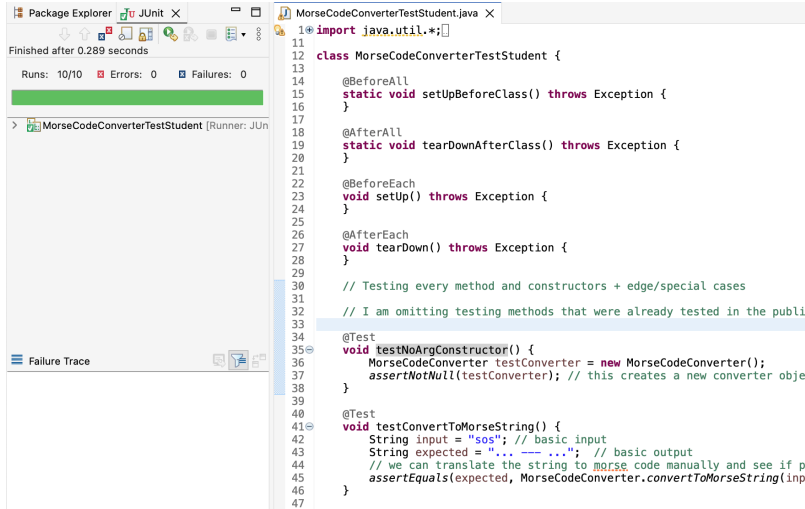
```
1
2 import static org.junit.Assert.*;
13
14
15 public class MorseCodeConverter_GFA_Test {
16
17     @Before
18     public void setUp() throws Exception {
19     }
20
21     @After
22     public void tearDown() throws Exception {
23     }
24
25     @Test
26     public void testConvertToEnglishString() {
27         String converter1 = MorseCodeConverter.convertToEnglish(".... . -.-. -.-. --- / .-- --- -. -.-. -.-. ");
28         assertEquals("hello world", converter1);
29     }
```

MorseCodeConverterPublicTest:

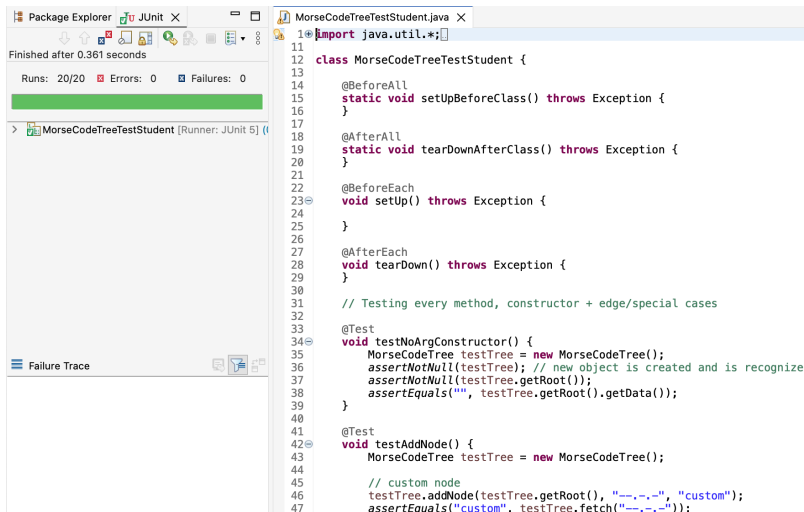


```
1
2 import static org.junit.Assert.*;
13
14
15 public class MorseCodeConverterTestPublic {
16
17     @Test
18     public void testConvertToEnglishString() {
19         String converter1 = MorseCodeConverter.convertToEnglish(".... . -.-. -.-. --- / .-- --- -. -.-. -.-. ");
20         assertEquals("hello world", converter1);
21     }
22
23     /**
24      * Testing for correct implementation of tree and traversal
25      */
26
27     @Test
28     public void testPrintTree() {
29
30         assertEquals("h s v i f u e l r a p w j b d x n c k y t z g q m o")
31     }
32
33     /**
34      * Testing for correct conversion of all characters using key phrase to
35      */
36
37     @Test
38     public void testConvertMorseStringToEnglishString() {
39
40         String converter1 = MorseCodeConverter.convertToEnglish("- .... . /
41         assertEquals("the quick brown fox jumps over the lazy dog", convert
42     }
43
44     @Test
45     public void testConvertMorseFileToEnglishString() {
46
47         /*Make sure howDoILoveThee.txt is in the src directory for this
48         test to pass
```

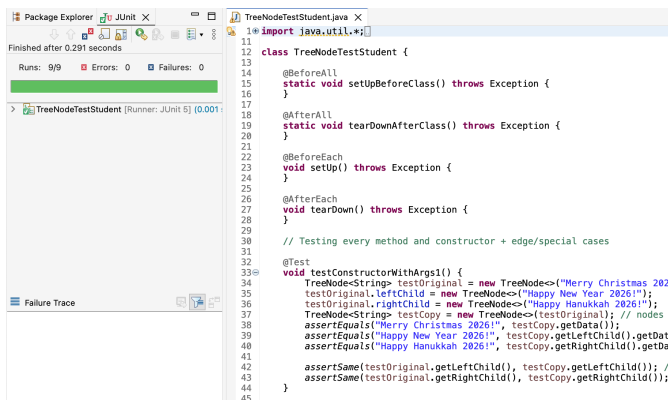
MorseCodeConverterTestStudent:



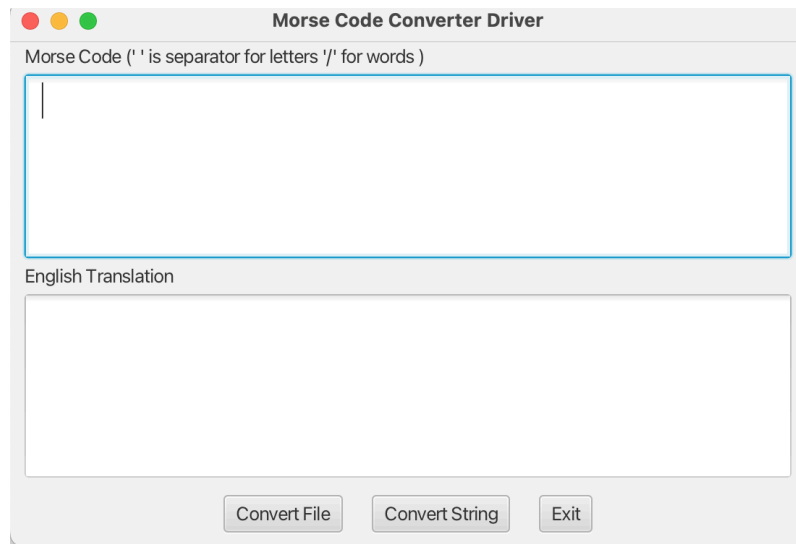
MorseCodeTreeTestStudent:



TreeNodeTestStudent:

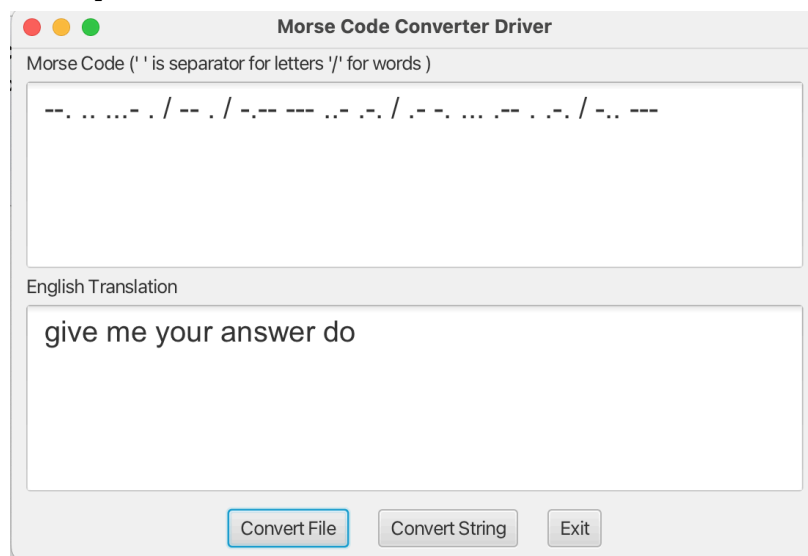


JavaFX GUI Screenshots: Initial Output (upon clicking "Run")



(all translations are in all lowercase)

Daisy.txt



DaisyDaisy.txt

Morse Code Converter Driver

Morse Code (' ' is separator for letters '/' for words)

.. -- / - .-. / -. -. -. -. -. -. -- / - .-. -. -. / -- . . / -
.... . / .-. -- .-. . / -- .-. / -. -- -. .-

English Translation

im half crazy all for the love of you

Convert File Convert String Exit

howDoILoveThee.txt

Morse Code Converter Driver

Morse Code (' ' is separator for letters '/' for words)

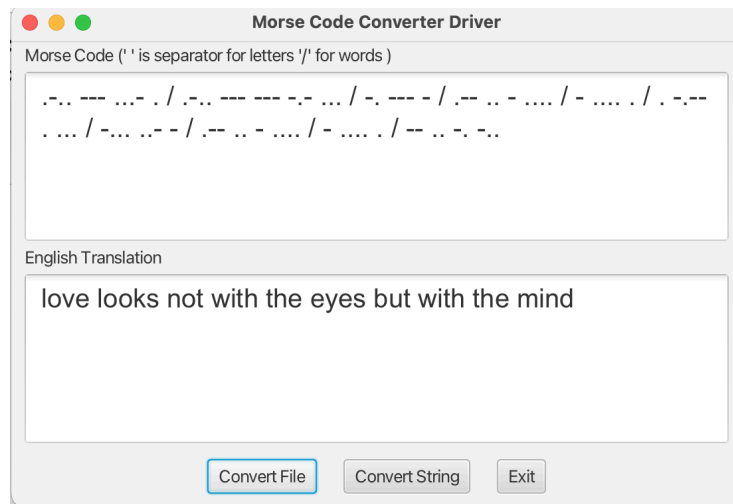
.... -- .- / -. -- / .. / .-. -- ...- . / - / .-. . - / -- . / -. .
-- .- - / - / .- .- -- ...

English Translation

how do i love thee let me count the ways

Convert File Convert String Exit

LoveLooksNot.txt



Stringtest

